

Camera Ready Summary

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International Conference on Electrical, Computer and Communication Technologies

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Paper Title

Predicting Student Academic Performance from Collaborative Learning Behaviors Using Machine Learning and Explainable Insights

Abstract

This work examines whether patterns of collaborative learning can be used to anticipate university students' academic performance through machine learning. The study draws on responses from 690 students collected via a Google Forms questionnaire. The survey captures both basic demographic information and a range of collaboration-related behaviors, resulting in a dataset of 13 features, including age, how often and how long students collaborate, typical group size, perceived helpfulness, improvement in understanding, perceived effects on grades and stress, motivation level, study preference, study location, and overall perception of collaboration.

Before model development, the data were preprocessed by converting categorical responses into numeric form, and class imbalance was handled using an oversampling approach. Multiple supervised learning algorithms were then trained and tested using a held-out portion of the data. Among the evaluated models, ensemble-based methods produced the most consistent results, with the top model reaching an accuracy of 0.977 and a ROC--AUC close to 0.99, indicating strong predictive performance.

An examination of feature contributions suggests that the level of engagement in collaborative activities and students' perceived learning benefits are particularly influential in predicting outcomes. Taken together, these results indicate that even lightweight, survey-based indicators of collaboration can support practical predictive models, which may be useful for early identification of performance patterns and for guiding targeted academic support strategies.

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